

DPhil Thesis Draft



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Abbreviations

CMOS complementary metal-oxide-semiconductor. 18

CNT carbon nanotube. 11, 12, 13

CNTFET carbon nanotube field-effect transistor. 12, 14, 16

DIBL drain induced barrier lowering. 14

EOT effective oxide thickness. 15, 16

FET field-effect transistor. 5, 12

MOSFET metal-oxide-semiconductor field-effect transistor. 8, 9, 10, 15

MW multi-walled. 11

SB Schottky barrier. 12

SS Sub-threshold swing. 14, 15, 16

SW single-walled. 11

1

Introduction

1.1 Motivation

Motivation

1.2 Graphene

1.2.1 crystal structure

Graphene has a hexagonal structure where the carbon atoms are hybridised sp^2 , as detailed by the Pauling theory on chemical bonds. Based on the experimental evidence collected in the last 20 years, the model that seems to best describe the proprieties of graphene is the band-structure Wallace model, which dates back in 1947, and describes graphene as a semimetal with a linear dispersion of the electronic excitations. In the International Table of Crystallography, graphene is included in the symmorphic space group #191, with symmetry D_6^1h ($P6/mmm$). The triangular lattice is obtained by considering two atoms per unit cell having the respective lattice vectors:

The carbon atoms in graphene form a two-dimensional hexagonal crystal with two atoms per unit cell, as shown in figure.

$$\mathbf{a}_1 = \frac{a}{2}(3, \sqrt{3}), \quad \mathbf{a}_2 = \frac{a}{2}(3, -\sqrt{3}), \quad (1.1)$$

with $a \approx 1.42$ Å be the carbon-carbon bond length. Each atom belonging to one sublattice, called sublattice A, has three first neighbours belonging to the other sublattice, called sublattice B. With respect to one atom chosen from sublattice A as origin, the position of these first three neighbours in real space are given by:

$$\boldsymbol{\delta}_1 = \frac{a}{2}(1, \sqrt{3}), \quad \boldsymbol{\delta}_2 = \frac{a}{2}(1, -\sqrt{3}), \quad \boldsymbol{\delta}_3 = -a(1, 0), \quad (1.2)$$

The reciprocal-lattice vectors are then given

$$\mathbf{b}_1 = \frac{2\pi}{3a}(1, \sqrt{3}), \quad \mathbf{b}_2 = \frac{2\pi}{3a}(1, -\sqrt{3}), \quad (1.3)$$

In order to define the vertices of the Brillouin zone one has to apply the Bragg condition:

$$2\mathbf{k} \cdot \mathbf{G} = 0 \quad (1.4)$$

where

$$\mathbf{G} = n\mathbf{b}_1 + m\mathbf{b}_2, \quad (1.5)$$

The first amongst these are:

$$\begin{array}{lll} n = \pm 1, & m = 0, & \mathbf{G} = \pm \mathbf{b}_1, \\ n = 0, & m = \pm 1, & \mathbf{G} = \pm \mathbf{b}_2, \\ n = \pm 1, & m = \pm 1, & \mathbf{G} = \pm(\mathbf{b}_1 + \mathbf{b}_2), \end{array}$$

and all these vectors have the same modulus $|\mathbf{G}| = \frac{4\pi}{3a}$. By applying Bragg's condition in the first of the three cases we get:

$$\sqrt{3}k_x + k_y = \mp \frac{4\pi}{3a}, \quad (1.6)$$

which gets satisfied by the points:

$$K \equiv \mp \frac{2\pi}{3a} \left(\frac{1}{\sqrt{3}}, 1 \right) \quad (1.7)$$

For the second case one gets:

$$\mp (\sqrt{3}k_x - k_y) = \frac{4\pi}{3a}, \quad (1.8)$$

which gives the points:

$$K' \equiv \pm \frac{2\pi}{3a} \left(\frac{1}{\sqrt{3}}, -1 \right) \quad (1.9)$$

The third and last case gives the point:

$$M \equiv \pm \left(0, \pm \frac{2\pi}{3a} \right) \quad (1.10)$$

1.2.2 band dispersion

The electronic structure of graphene can be model with a tight-binding hamiltonian. Following the second quantization formalism ($\hbar = 1$ units), the hamiltonian can be written as:

$$\mathcal{H}_{\text{TB}} = -t \sum_{\langle ij \rangle, \sigma} \left(a_{\sigma,i}^\dagger b_{\sigma,j} + \text{H.c.} \right) - t' \sum_{\langle\langle ij \rangle\rangle, \sigma} \left(a_{\sigma,i}^\dagger a_{\sigma,j} + b_{\sigma,i}^\dagger b_{\sigma,j} + \text{H.c.} \right), \quad (1.11)$$

The notation $\langle ij \rangle$ means that the sum runs over pairs of electrons which are nearest neighbour; similarly, the notation $\langle\langle ij \rangle\rangle$ specify a summation over the next nearest neighbours; the operators $a_{i,\sigma}^\dagger$ ($a_{i,\sigma}$) and $b_{i,\sigma}^\dagger$ ($b_{i,\sigma}$) creates (annihilates) an electron with spin σ ($\sigma = \uparrow, \downarrow$) for atoms on sublattice A and B, respectively. The value of t which reproduces best experimental result is $t \approx 2.8$ eV and quantify the nearest-neighbour hopping energy; similarly, t' denotes the next nearest-neighbour hopping energy. The dispersion energy computed from this model reads:

$$E_{\pm}(\mathbf{k}) = \epsilon_p - t' f(\mathbf{k}) \pm t \sqrt{3 + f(\mathbf{k})}, \quad (1.12)$$

with $f(\mathbf{k})$ defined as:

$$f(\mathbf{k}) = 2 \cos(\sqrt{3}k_y a) + 4 \cos\left(\frac{\sqrt{3}}{2}k_y a\right) \cos\left(\frac{3}{2}k_x a\right). \quad (1.13)$$

Therefore, it is possible to identify an upper (π^*) band associated with the plus sign and a lower (π) band described by the minus sign; these are symmetric with respect to the single-particle zero energy. The experimental estimate reported for next nearest-neighbour hopping energy is $t' \approx 0.1\text{eV}$ thus the asymmetry introduced is small. It is worthwhile to notice that the function $3 + f(\mathbf{k})$

If the system is not doped, there is one electron per carbon atom, so two electrons per cell: it follows that the lower band is full and the upper band is empty the upper one is empty. The system is therefore a semi-metal. The peculiarities of the dispersion in graphene become apparent when analysing their trend near the point $\mathbf{K}$ and $\mathbf{K}'$. By taking the positive components for the $\mathbf{K}$ as define in Eq. 1.7 and expanding the Eq. 1.12 as $\mathbf{k} = \mathbf{K} + \mathbf{q}$, with $|\mathbf{q}| \ll |\mathbf{K}|$, one gets:

$$E_{\pm}(\mathbf{q}) = \epsilon_p \pm \frac{3at}{2}|\mathbf{q}| \quad (1.14)$$

which is linear in $\mathbf{q}$. For intrinsic graphene, at $T = 0\text{K}$ $E_F = \epsilon_p$. The Fermi velocity is defined by the usuale relation $v_F = \frac{1}{\hbar} \left(\frac{dE}{d\mathbf{k}} \right)$, and in graphene $v_F \simeq 1 \times 10^6 \text{m s}^{-1}$.

A difference arises in the density of the states, when we limit ourselves to consider those near the Fermi energy. In the case of a two-dimensional electron gas, it is constant. On the contrary for graphene, the density of states per unite cell is:

$$\rho(E) = \frac{2A_{\text{cell}}}{\pi} \frac{|E|}{v_F^2} \quad (1.15)$$

If the graphene is charge-neutral, the DOS has two Van Hove singularities at $\omega_{\text{VH}} = \pm 1\text{eV}$ [1]. This behaviour is in stark constrat with the DOS of carbon nanotubes and clean graphene nanoribbons, as they show $1/\sqrt{E}$ singularities due to the 1D nature, which imposes a quantization perpendicular to the ribbon or tube longitudinal direction.

1.2.3 Graphene Transistors

For transistor applications, a material needs a highly conductive on-state and an off-state where the source-drain current I_{sd} does not flow. For RFs applications, on the other hand, the full speed operation is obtained by operating in the (drain) current saturation regime. Both applications requires a sizeable band gap. Furthermore, the IC scaling trend is often juxtaposed to chip-level issues such as power densities (currently on the order of 1000 W/cm^2), heat generation, and chip temperatures reaching levels that will prevent reliable operations.

In CMOS logic, the charge carriers are accelerated by the electric field $\mathcal{E}_{sd}$ between source and drain, thus undergoing five possible sources of scattering: electron–electron scattering, electron–phonon (lattice vibrations), material interfaces, imperfections or impurity atoms. The first two are more fundamentals, while the last three are material-related. The most dominant mechanism through which charge carriers lose net energy is by scattering with phonons, consequently heating up the lattice via Joule heating. For RF applications, on the other hand, heat is produced by the operational frequency at which the carriers operate. Thus, two further requirements are needed: on one hand a good thermal conductivity, but also small heat transfer with the substrate. Clearly trade-offs are here necessary.

The 2020 International Roadmap for Devices and Systems - which has replaced since 2017 the International Roadmap for Semiconductors - dedicates ample attention to graphene in its report *Beyond CMOS*, in which technologies and semiconductor materials to complement and replace Si are considered. The race to scale the channel in a field-effect transistor (FET) still motivates a significant effort on introducing high-mobility channel materials, since the carrier mobility is an appropriate parameter to address carrier transport even in nanometer FETs [2]. In fact, most of the early excitements in using graphene as channel material in a standard FET technology were sparked by considering the outstanding mobility values reported since its discovery.

The first data were the one obtained in 2004, where graphene on SiO_2 displayed an electron mobility $\mu_e \approx 10,000 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}$ [3]. Later in 2008 [4] and 2013 [5], higher

mobility values were reported: $\approx 120,000 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}$ at 240 K and $\approx 1,000,000 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}$ at 4 K. The reason for these higher values was attributed to the use of suspended graphene rather onto substrate, where surface effects inevitably affect the carrier dynamics. At the same time, results ($\approx 120,000 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}$) obtained at room temperature with graphene onto a single-layer of hexagonal boron nitride (hBN) [6] and sandwiched between two hBN layer [5] have suggested that this flat and inert materials can be act as a excellent passivation film for graphene devices.

One of the figure of merit often taken into account for addressing logic applications for a materials is the ratio between the on-state current and the off-state current. The so called on-off ratio $I_{\text{on}}/I_{\text{off}}$ has to be in the range 10^4 – 10^7 . While the I_{on} is not directly related to the band gap E_g , it is possible to link the value of I_{off} with the E_g .

In an metal–semiconductor junction (M/S), the current is built by the majority of the carriers, these being electrons or holes for n -type or p -type semiconductors. Most of the graphene–semiconductor junctions formed during experiments have typical dopant concentration $N \leq 10^{17} \text{ cm}^{-3}$. The $I - V$ characteristics of M/S junctions with such dopant concentrations are well reproduced by the thermoionic emission theory. If we then exclude other possible conduction mechanisms (standard and thermoionic field emission, generation/recombination, diffusion of electron, injection of holes and leakage current), the I_{off} becomes proportional to the barrier the electrons have to overcome from the semiconductor to the metal, namely

$$I_{\text{off}} \propto \exp\{(e\beta\phi_B)\}. \quad (1.16)$$

where e is the electron charge, ϕ_B is the Schottky barrier height, and $\beta = k_B T$. If we assume that the work function of graphene and the metal are the same, then $\phi_B = E_g/2$, where E_g is the band gap energy originating from the difference of the work function of graphene and the metal; thus, the on–off ratio can be estimated from the useful proportionality [7]:

$$\frac{I_{\text{on}}}{I_{\text{off}}} \propto \exp\left\{\frac{E_g}{2k_B T}\right\}, \quad (1.17)$$

	Direct Gap			Indirect Gap		
	InP	Si	Ge	GaAs	InAs	Graphene
a_{lattice} (Å)	5.87	5.43	5.65	5.65	6.06	2.46
E_g (eV)	1.35	1.12	0.66	1.42	0.36	0
μ (cm ² V ⁻¹ s ⁻¹) at 300 K	2800	1400	3900	4600	16,000	200,000
v_{sat} (10 ⁷ cm s ⁻¹)	2.2	1	0.6	2.2	4.0	>5
DOS (m^*/m_0)	0.077	1.08	0.56	0.067	0.023	0
ϵ_r	12.4	11.9	16.0	13.1	14.6	2.4
κ (W m ⁻¹ K ⁻¹)	68	150	60.2	46	27	5000

Table 1.1: Differences between the main semiconductors used in modern digital electronics and graphene.

where the thermal energy $k_B T$ and the bandgap are both expressed in meV. While for high frequency applications the on–off ratio is of minor importance, several references in literature report suitable values for CMOS bandgap should fall in a range between 300 and 500 meV [8, 9].

Hysteresis between the forward and reverse sweeps have been reported depending on the sweeping rate and range [10], and attributed mainly to charge trapping in silanol groups (Si–OH), mediated and assisted by surface-bound H₂O molecules [11]. Despite the usefulness for its proof-of-concept purposes, back-gate devices suffered from parasitic capacitances, making impossible further integration with other components. Alternatives geometries have since then been explored, in order to increase the carriers mobility, to decrease contact resistances and parasitic capacitances, and to avoid Schottky barriers formation between the graphene and the semiconductor. Top-gate devices have been the first to chronologically appear in the literature. Back then in 2007, Lemme and co workeres used a stack of SiO₂ (20 nm), Ti (10 nm), and Au (100 nm) as top gate for modulating the current

inside a large-area graphene channel transistor obtained by the exfoliation method [12]. Whilst the gate characteristics exhibit a similar trend of the characteristics (the top gate reduces the back-gate current modulation, lowering the drain current by one order of magnitude) and the hysteresis, weak current saturation at high drain-source bias have been reported [13]. This effect seem due to a partial charge pinning at the contacts at high back-gate voltages [14] and carrier drift velocity eventually saturating due to optical-phonon scattering [15]. Top-gated graphene metal–oxide–semiconductor field-effect transistors (MOSFETs) with CVD and epitaxial graphene have been also made [16, 17]; SiO₂, Al₂O₃, and HfO₂ have been used as top-gate dielectric.

However, despite the improvements obtained with a top gate, most graphene MOSFET with large-area graphene channels show on–off switching ratios of less than 10 at room temperature [8], much too little for complex logic circuits.

In 2012, Mehr *et al.* proposed a vertical arrangement of emitter, base, and collector similar to the original vacuum triode, where graphene plays the base role and electrons travel vertically rather than transversely. When the device is in the ON state, the electrons tunnel with a Fowler-Nordheim tunnelling mechanism:

$$I \propto \left(\frac{V_{sd}}{d} \right)^2 \exp \left\{ -\frac{\alpha d}{V_{sd}} \right\}, \quad (1.18)$$

through the emitter–base layer (of thickness d); ON/OFF ratios of the order of 10⁴ have been demonstrated using graphene between two h-BN layers [18] and using WeS₂–graphene [19] heterostructures.

Solution methods have also been explored, in which graphene is dropcasted on top of a semiconducting wafer, and then spin-coated [20]. The main difficulties arise when attempting to quantify the structure and the quality of the deposited graphene, and not many methods can guarantee a good dispersion in organic solvents [21]. Simpler ways to obtain graphene oxide in solution have been reported [22], but once deposited, graphene oxide needs to be reduced for obtaining pure graphene – low mobility values (up to, at best, 1000 cm² V^{−1} s^{−1}) caused by defects during the fabrication steps have discouraged further investigations along this direction.

Yet in spite of the very different dc behavior compared to the most used semiconductors in digital electronics, graphene FETs have reported good performance in terms of high-frequency applications. This comparison is made by measuring each system's frequency response, its output signal strength when it is given a particular input signal: features such as ballistic transport, linear current-voltage characteristics, high transconductance, and very high ($> 10^8 \text{ A cm}^{-2}$) sustainable currents are all ideal features for analog/RF applications [15, 23].

The primary means of estimating frequency response is through the evaluation of two transistor characteristic frequencies, namely cutoff f_T and maximum of oscillations f_{max} . The former indicates how quickly a signal can travel from the gate to the drain of the device, and it is usually determined when the small-signal current gain drops to unity, while the latter is measured when the power gain decreases to unity [8]. The frequency response depends inversely on gate length and proportionally to the transconductance, which is loosely proportional to mobility. Thus, reducing the gate length and using higher mobility materials are valid approaches to increase the device operational speed [24]. In this context, many research groups have successfully demonstrated RF capabilities with gapless graphene channels MOSFETs.

The first device with cutoff frequency f_T in the GHz range was reported in late 2008, and was fabricated by exfoliated graphene deposited on top of high-resistivity Si substrate, with 30-nm-thick top-gate HfO_2 dielectric and 500 nm of gate length [23]. Despite a measured $f_T \approx 14.7 \text{ GHz}$, it was argued that large contact resistances and gate length were the main responsible of poor performance $f_{\text{max}} < 1 \text{ GHz}$.

The following year, Pd/Au contacts and 12-nm-thick Al_2O_3 were employed in a 350-nm-gate [25]. While the cutoff frequency improved up to $f_T \approx 50 \text{ GHz}$, little gains in terms of f_{max} were obtained of only 2 GHz. In general, limited performance in f_{max} represents a major issue as for most RF applications f_T is less important than f_{max} .

It was only after two years that an increase of one order of magnitude both in the cutoff and the maximum frequency of oscillation was demonstrated [17]. The device

was realised with epitaxially-grown on silicon carbide (SiC) graphene MOSFET having 240 nm of gate length, with Ti/Pd/Au for contacts and 10-nm-thick HfO₂ of insulating layer on top gates. Highest frequency as well as $f_T \approx 100$ GHz and $f_{\max} \approx 14$ GHz were measured, exceeding any device then present. The authors also argued that since $f_{\max}$ is strongly correlated to device layout and parasitic resistances originating from the metallic contacts, viable ways to improve this parameter would be a partial passivation of the metal contacts and a multi-finger gate layout

These performance were soon followed later that year by a 144-nm-length gate device with $f_T \approx 300$ GHz using mechanically exfoliated graphene and Co₂Si–Al₂O₃ nanowire as self-aligned multi top-gates [26]. This result was also corroborated by comparing it with the projected value using the relation that links dc transconductance and gate capacitance with the cutoff frequency when total input parasitic capacitance is negligible, namely $f_T = g_m/(2\pi C_g)$ [27]. Unfortunately, the authors did not report the power gain $G_{\max}^{1/2}$ but only the current $|h_{21}|$, thus is not clear whether this technology had moved forward in terms of $f_{\max}$.

Finally, in 2012, a graphene MOSFET with a 40-nm-long gate showing 350-GHz f_T [28] and a 67-nm gate graphene MOSFET with an f_T of 427 GHz have been reported [29]. Graphene RF MOSFETs have successfully been realized using exfoliated (e.g., [26] and [25]), epitaxial (e.g., [17] and [13]), and CVD ([30]) graphene. Fig. shows the cutoff frequency of the best graphene MOSFETs reported, together with the f_T performance of competing RF FET types. The record f_T data of the competing FETs are 688 GHz for a 40-nm GaAs mHEMT[31] and 644 GHz for a 30-nm InP HEMT [32], 485 GHz for a 29-nm Si MOSFET [33], and 153 GHz for a 100-nm carbon nanotube MOSFET [34]. Worthwhile to mention also the results obtained by Han *et al.* [35] in 2014, when they implemented a high-performance three-stage graphene integrated circuit using Si fabrication steps and CVD graphene: the reported performance for f_T and $f_{\max}$ were ranging from 8 to 10 GHz and from 8.5 to 13.5 GHz.

1.3 Carbon-based Field Effect Transistor

1.3.1 crystal structure

This crystalline form of carbon can be considered as a rolled up sheet of graphene and comes in two major flavours: single-walled (SW) and multi-walled (MW). Carbon nanotubes (CNTs) display an electronic band gap which depends on the chirality and the diameter of the nanotube [36]. Having defined two unit vectors in the real space, $\mathbf{a}_1$ and $\mathbf{a}_2$, a CNT can be described by the linear combination of these with integers coefficients (m, n) . Based on the values of m and n , CNTs can be metallic, quasi-metallic ($m - n$ is a multiple of 3, with $m \cdot n \neq 0$ and $m \neq n$), and semiconducting for the majority of the remaining cases – exceptions due to the curvature (spin-orbit couplings) in small-diameter tubes are also known [37]. Thus, a quantization of the electronic states in the circumferential direction is resulted when the nanotube diameter d_{CNT} is between 0.8 and 2.5 nm. Each of these substates is called subband and are determined by the periodic boundary condition. Having defined a circumferential wave vector $\mathbf{k}_c$ such that $\mathbf{k}_c \cdot \mathbf{c} = k_c |c| = k_c \pi d_{\text{CNT}} = 2\pi\nu$, for ν integer, it is possible from this relation to draw parallel lines in the first Brillouin zone of graphene. If the subband intersect one of the $\mathbf{K}$ points, then the resulting nanotube will have no energy gap and thus exhibit metallic properties (this condition is realised by having $m = n$ for the coefficients of the chiral vector). If none of the subbands pass through $\mathbf{K}$ then the resulting CNT will be semiconducting, having a E_g proportional to the smallest distance between a $\mathbf{K}$ point and the subband lines:

$$E_g = 2 \left(\frac{\partial E}{\partial k} \right) \frac{k_c}{3} = \frac{2}{3} \hbar v_F \frac{2\pi}{|c|} = \frac{4}{3} \hbar v_F \frac{1}{d_{\text{CNT}}} \approx \frac{0.7}{d_{\text{CNT}}} \quad (1.19)$$

where $v_F = 8 \times 10^5 \text{ m s}^{-1}$ is the Fermi velocity, E_g is expressed in eV, and d_{CNT} is expressed in nm.

1.3.2 Carbon Nanotube Field-Effect Transistors

General Operation

The modulation of a CNT channel by a gate-induced field was realised in the 1998 by the group of Cees Dekkar. These early FETs were fabricated by using doped silicon substrate as a gate to modulate the energy bands, separated from the nanotube by thermally-grown SiO_2 oxide acting as dielectric. For contacting the nanotube, metal contacts were employed.

Since then, a great variety of carbon nanotube field-effect transistor (CNTFET) structures and geometries have been demonstrated, and can be essentially classified in devices with or without a spacer. where the spacer length L_{spr} is the distance between the gated channel and the source/drain metal contacts. Given that the nanotubes are intrinsically semiconducting, in spacer-less devices charge carriers are not induced in the channel by the gate-field but are injected from the contacts: the role of the gate is to lower the thermal barrier in the channel, thus allowing a tunnelling through the Schottky barrier (SB) at the source. In this scenario, a critical role for the CNTFET performance is played by the metal work function ϕ^{M} .

CNTFETs with spacers, on the other hand, are less limited by the injection at the junction source–CNT, as they are regulated by the modulating effect played by in the channel as in conventional. Since the carrier barrier injection at the drain is large, device with spacers have been proved to be more resistant to ambipolar conduction. In Table 1.2 an overview of gated-spacer (top and bottom) devices is reported.

Contacts

One of the principal issues when study the interface metal–CNT is the role played by carbon and metal atoms at the interface. Solid-state semiconductors have dangling bonds at the surface, leading to Fermi level pinning, meaning that the variation of the band bending is independent of the work function of the metal, even when this is large. Since the ideal structure of CNTs has no open bonds, no pinning of the metal Fermi level to mid-gap states occurs, thus legitimating the study of different work functions for maximising performance and tuning polarity. Many metals have

<i>Device type</i>	$I_{\text{on}}/I_{\text{off}}$	SS (mV/dec)	g_{mt} ($\mu\text{S cm}^{-1}$)	L_{ch} (nm)	f_T [f_{max}] (GHz)	Ref.
Bottom gate	10^4	94	~ 40	9	—	[38]
	10^5	~ 85	40	15	—	[39]
	10^5	~ 85	20	20	—	[39]
	10^5	60	20	200	—	[40]
	10^5	~ 63	—	200	—	[41]
	10^4	80	50	240	—	[42]
	—	—	143	800	[1]	[43]
	10^4	—	—	1,500	1.5 – 3.0	[44]
	—	70	$12^\dagger$	2,000 ¹	—	[45]
	10^8	—	—	$< 10,000^1$	—	[46]
	—	—	- 500	—	30	[47]
Top gate	10^6	—	20^2	80	—	[48]
	2	—	~ 50	100	25	[49]
	50	—	310	100	~ 70 [80]	[50]
	—	—	0.3	210	—	[51]
	10^5	—	33.5	85,000	—	[52]

Table 1.2: Figures of merit for a few CNFET reported in literature. S : subthreshold swing; g_{m} : intrinsic transconductances; L_{ch} : channel length; μ : carriers mobility

[†]Normalized by the tube width.

¹ Referred to the gated length of a nanotube.

² For the n-type transistor, whereas 10 μS were measured for the p-type.

<i>Device type</i>	I_{on} (A)	I_{off} (A)	$I_{\text{on}}/I_{\text{off}}$	g_{mt} (μS)	SS (mV/dec)
Conventional	1.41×10^{-5}	4.06×10^{-10}	3.4×10^4	29	99.6
Partial S/D-doped	1.50×10^{-5}	2.33×10^{-8}	6.4×10^2	63.6	93.1
Schottky-Barrier	3.26×10^{-6}	2.31×10^{-8}	1.41×10^2	19.6	109.8
Tunnel	5.53×10^{-6}	1.16×10^{-8}	4.76×10^2	13	107.9

Table 1.3: Comparative simulation for the four major type of CNFETs as reported in [53]. These results are obtained by setting $V_{\text{sd}} = 0.5$ V.

been studied for contacting CNTs. Interestingly, it has been demonstrated that Pd ($\phi^{\text{Pd}} = 5.22 - -5.60$) builds an almost ideal p -channel contact to the valence band of CNTs. Despite having a larger workfunction ($\phi^{\text{Pt}} \approx 5.65$ eV) than Pd, Pt provides very poor contacts due to the physical wetting of the metal on the nanotube.

***n*-Type CNTFETs**

In a *n*-type CNTFETs the injection of electrons is favoured over the holes. Given that $\phi^{\text{CNT}} \approx 4.7$ eV and with an average $E_g \approx 0.6$ eV, it is possible to configure *n*-type channel with nanotube by either doping the spacer region or by using a metal with lower work function. One of the main drawbacks of doping is the challenge to keep it under control: instability in air, low consistency, variability to temperature variations, are amongst the most frequent scenarios. The other issue in working with low work function metals as contacts is the high tendency to oxidize. A simple and effective workaround to this is to use high work function metals to cap the contact metals, passivating the contact surface tendency to destabilise in air. One example of high performance *n*-type device was obtained by using Y contacts, achieving an Sub-threshold swing (SS) of 73 mV/dec, mobility of $5100 \text{ cm V}^{-2} \text{ s}^{-1}$, and drain induced barrier lowering (DIBL) of 105 mV V^{-1} [54].

Scaling

While Si technology was for long time questioned whether it could work below 10-nm of channel length, recent impressive technological developments from the industry have proved not only its feasibility but also a widespread commercial application (currently modern laptop employ the 5-nm technology). Nevertheless, the interest in Si scalability has been always a crucial starting point for exploring different technologies and materials capable of overcoming short-channel effects (threshold voltage roll-off, off-state leakage current, channel length modulation are just a few of the problematics).

In carbon nanotubes transistors a crucial and important – but somehow overlooked difference – is the difference between the two critical lengths for scaling, which are the contact length (L_c) and the channel length (L_{ch}). Aggressive scaling for the channel length generally leads to lose control of the band bending through the gate: intuitively, by shrinking the channel length the electric field generated by the source contact will be felt more and more by the charge carriers. The argument here is that carbon nanotubes offer, however, a good enough platform for such

scaling given by the ultrathin hollow structure (recall an effective wall thickness about $d_{\text{CNT}} \approx 0.68 \text{ \AA}$). On the other hand, this could be a double edge-sword. The effective carrier mass in nanotubes is $m_{\text{CNT}}^* = (0.228 \pm 0.129)m_0$ [55], thus potentially leading to tunnelling current from source to drain: this effect can be appreciate by an increase in the SS and the OFF current.

One of the first few systematic study on channel length scaling effects was conducted by Franklin *et al.* in 2010 through a series of measurements on nanotubes with relatively small diameters ($1.1 \text{ nm} < d_{\text{CNT}} < 1.3 \text{ nm}$) by using local bottom gates covered with 2 nm effective oxide thickness (EOT) of HfO_2 [56]. A consistent trend was observed when the channel length was scaled down more than two orders of magnitude (from 3 μm to 15 nm), both in the values of V_{th} and SS ($-0.5 \pm 0.2 \text{ V}$ and 90 mV/dec, respectively). The authors reported also encouraging figures on the transconductance g_m .

In a standard MOSFET, the transconductance is the change in the drain current divided by the amount varied in the gate/source voltage V_{gs} when the drain/source voltage (V_{ds}) is kept constant. Typical values of g_m for a small-signal field effect transistor are between 1 and 30 $\text{mS}\mu\text{m}^{-1}$. In a n-channel (nMOS) FET with threshold voltage V_{th} , the drain current at a certain V_{gs} in saturation mode (so when $V_{\text{th}} > V_{\text{gs}} - V_{\text{sd}}$) is given by:

$$I_{\text{sd}} = \frac{1}{2}\mu_n C_{\text{ox}} \frac{W}{L} (V_{\text{gs}} - V_{\text{th}})^2, \quad (1.20)$$

where μ_n is the electron mobility, C_{ox} is the gate oxide capacitance, W and L are the gate width and length, respectively. It is possible to work out the value of g_m by using the Shichman-Hodges model and assuming that the threshold voltage V_{th} is a device constant, such that $V_{\text{ov}} = V_{\text{gs}} - V_{\text{th}}$:

$$g_m = \frac{\partial I_{\text{sd}}}{\partial V_{\text{gs}}} = \frac{2 I_{\text{sd}}}{V_{\text{ov}}}, \quad (1.21)$$

Here V_{ov} is the so called *overdrive voltage* and has the physical meaning of being the voltage above the threshold voltage of the MOSFET. It is useful in the designing

phase of a transistor and elegantly links the drain current with a voltage: in fact, by rearranging Eq. 1.20 and identify $k_n = \mu_n C_{\text{ox}} \frac{W}{L}$ one gets:

$$I_{\text{sd}} = \frac{1}{2} k_n V_{\text{ov}}^2 \implies V_{\text{ov}} = \sqrt{\frac{2I_{\text{sd}}}{k_n}} \quad (1.22)$$

The usefulness for this equation is that the relation expressed in terms of overdrive voltage rather than threshold voltage holds both in the n and in the p case. Some reference numbers: V_{ov} is typically set at about 70–200 mV for the 65 nm technology node (with $I_{\text{sd}} = 1.13 \text{ mA } \mu\text{m}^{-1}$ of width) with value of $g_{\text{m}} \approx 11 - 32 \text{ mS } \mu\text{m}^{-1}$. In their work, Franklin *et al.* achieved transconductance peak up to $40 \text{ } \mu\text{S } \mu\text{m}^{-1}$ for a CNTFET with 15 nm of channel length, corresponding to values of g_{m} between approximately $91 - 110 \text{ } \mu\text{S } \mu\text{m}^{-1}$, thus meeting the requirements for being compatible with the Si technology.

The first CNTFET in which L_{ch} was scaled below 10 nm came two years later, again by Franklin *et al.* [38]. By using a CNT with $d_{\text{CNT}} \approx 1.3 \text{ nm} \pm 0.2 \text{ nm}$ and $E_g \approx 0.62 \text{ eV} \pm 0.1 \text{ eV}$ contacted through local bottom gates, they manage to scale down the $L_{\text{ch}} = 9 \text{ nm}$. The EOT was reduced to $\approx 0.65 \text{ nm}$ to achieve a better electrostatic coupling with the nanotube.

The authors reported an increase on the minimum current as L_{ch} decreases from 320 nm to 9 nm, attributed to a back-injection of the carriers into the conduction band from the drain, while the on-current approached the ballistic value for $L_{\text{ch}} \leq 41 \text{ nm}$ [56]. Impressive switching behaviour, with SS about 94 mV/dec, was measured, thus far outperforming any silicon-based of that time. based on previous results [56], the authors claimed the possibility of lowering the SS and the off-current at minimal cost to I_{on} by reducing the diameter of the nanotubes.

1.4 Graphene Nanoribbons

band structure

Historically, a first classification of graphene nanoribbons was derived by grouping the edge structures into Armchair-GNRs (AGNRs) and Zigzag-GNRs (ZGNRs).

Following the success of chemical routes in producing GNRS, various kind of edges have come to light, e.g., cove, fjord, chiral, and all these can be regarded as combination of AGNRs and ZGNRs [Figure](#).

Initial bandgap estimations through DFT calculations reported values between 2.5 and 0.5 eV. The inclusion of electronic correlation, however, becomes crucial for ultra-narrow (< 5 nm) ribbon [57, 58] and more accurate estimates derived by the group of Steven Louie based on the *GW* approximation reported a wider range of energy gaps from 5.5 to 1 eV. These calculations have shown that, as rule of thumb, AGNRs can be both semiconducting and metallic, while ZGNRs should be regarded as quasi-metallic, with energy gaps lying within 0.1 - 0.7 eV. Indeed, for this kind of ribbon the application of an electric field in the transverse direction of induces a spin-polarisation of the edge states is believed to be responsible for semiconducting properties of ZGNRs. The simplest model for describing the energy levels for a graphene nanoribbon is the particle in the box model, where the energy should scale quadratically $E_g \propto 1/L^2$ with the ribbon extension. Experimentally, however, the band gap for AGNRs scales with the inverse of the width $E_g \propto 1/W$ [57]: by considering all the interatomic carbon-carbon distances the same, it is possible to work out an averaged formula which reports the band gap varying as $E_g = 0.8/W$, [59]. More accurate simulations have shown, however, that peripheral carbon—carbon bond lengths usually tend to be larger compared to the ribbon core ones [60]. It is predicted that the band gap will be similar to Ge or InN when the width is reduced to 2 and 3 nm, and will be similar to Si or GaAs when it is further reduced to 1-2 nm [61].

AGNR

For AGNRs, the bandgap varies significantly for different m values. AGNRs exhibit metallic behavior when $m = 3p + 2$, where p is an integer number, and they become semiconductors when $m = 3p$ or $3p + 1$. These relations have been obtained through *in silico* experiments, both using the tight-binding method and DFT. Figure 3 presents the isosurface plots of Highest Occupied Molecular Orbital (HOMO) and

Lowest Unoccupied Molecular Orbital (LUMO) of AGNRs with $m=7,8,9$ respectively. Interestingly, the HOMO and LUMO orbitals are localized at the terminal of the ribbon along the length direction when $m=7(3p+1)$.

ZGNR

The band structure of ZGNRs is more interesting. Tight binding estimations have shown that if spin DOFs are excluded, the conduction band and valence band touches at the fermi level, forming a flat band. This flat band leads to a large localization of the density of states at the edges of ZGNRs, the so called *edge states*, which displays an exponential decay $\rho(\epsilon \sim \epsilon^{-r})$, with an exponent depending of the distance from centre of the ribbon; the weight of these states become negligible when the width w is the order of μm . The formation of band gap is observed when the spin is considered. In particular, the ground state of ZGNRs has parallel (ferromagnetic, $\uparrow\uparrow$, FM) spin alignment along one edge, whilst two edges have antiparallel (antiferromagnetic, $\downarrow\downarrow$, AFM) configuration. Given that the AFM state is semiconducting while the FM is metallic, it would theoretically be possible to change the metallic state of a zigzag ribbon by applying a magnetic field in the direction transverse to that of the ribbon's extension.

The scientific interest for graphene nanoribbons started in the physics community. Although this may appear irrelevant at the first glance, we will see that this actually had an impact over the last twenty years of progress in the field. The physics community was looking for a way to open a band gap in graphene. As mentioned in Sec. 1.1, large-area graphene is a semimetal, therefore unsuitable for complementary metal-oxide-semiconductor (CMOS) logic and problematic for RF.

The first class of methods for synthesising graphene nanoribbons are methods that involve cutting or producing rectangular shapes from the graphene lattice through macroscopic methods. For this reason, they are also known as *top-down* approaches. Although they have seen an improvements in recent years, common

drawbacks for this class of methods are a random distribution in the widths, a non-atomical control of the edge structures, and non-negligible impurity concentrations left after the process.

Kim and co-workers synthesized GNRs by employing the electron-beam lithography and etching technique [62]. The widths of GNRs obtained by this technique were scattered over a broad range, and most of the edges were extremely rough. An 8 nm GNR FET showed a dramatically improved on/off ratio of about 160.

Nanocutting of graphene is another promising method of GNR synthesis. In 2008, Strachan *et al.* used thermally activated Fe nanoparticles as chemical scissors to cut graphene sheet along a specific crystallographic direction [62, 63]. Later, Ni and other catalysts were employed as well. The carbon-carbon bond breaking is driven by the combustion reaction $\text{Ni} + \text{C}_{\text{graphene}} + 2\text{H}_2 \uparrow \longrightarrow \text{Ni/Fe} + \text{CH}_4 \uparrow$ catalysed by the nanoparticles, thus producing hydrocarbon compounds. One of the reasons that have discouraged these synthetic ways is the random width distribution obtained: this is likely to be a consequence of the change in direction when nanoparticles encounter defects or edges.

edge states

Band gap opening and magnetic ordering are two main features of ZGNRs when the electron-electron correlations and the spin DOFs are considered. Steven Louie's group extended the computation also to chiral edges, elucidating the inherent role of spin-polarized magnetic edge states in these kind of ribbons. [64]. They employed a mean-field Hubbard model to calculate the band structure, assuming the magnitude for the on-site Coulomb repulsion being comparable to the hopping integral $U/t \sim 1$. The Hubbard hamiltonian using the second quantisation formalism resembles Eq. 1.11 and can be expressed as:

$$\begin{aligned} \mathcal{H}_{\text{MF}} = & -t \sum_{\langle ij \rangle, \sigma} (a_{\sigma,i}^\dagger a_{\sigma,j} + \text{H.c.}) \\ & + U \sum_i (n_{i\uparrow} \langle n_{i\downarrow} \rangle + \langle n_{i\uparrow} \rangle n_{i\downarrow} - \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle), \end{aligned} \quad (1.23)$$

except for the second term, which is given by the summation of spin-polarised density $n_{i\sigma} = a_{i\sigma}^\dagger a_{i\sigma}$ and their expectation values. The hamiltonian in Eq. 1.23 respects the electron-hole symmetry, meaning that the density of states is symmetric respect to the E_F . The solution of the model displays interesting features: two pairs of peaks split (Δ_0 and Δ_1) in the density of states are observed, one associated with the electronic band gap, and one associated with the maximum energy splitting. The authors claim that the first peak split arises due to an AFM correlation between opposite edges, while Δ_1 is a consequence of a ferromagnetic order due to strong localised state along a single edge. Within this description, it is legit to link the behaviour of chiral GNRs with the one of spin-polarised ZGNRs; thus the total energy difference between ferromagnetic and antiferromagnetic couplings should be in the order of tens of meV per edge atom and should decrease as the width of the ribbon increases, eventually becoming negligible when the ribbon width is significantly larger than the decay length of the spin polarized [65]. Both features are in fact recovered from the solution of Eq. 1.23.

2

Foundations

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1.4	Graphene Nanoribbons	16

2.1 Charge Transport In Zero-Dimensional Objects

quantum dots transport physics

2.2 Charge Transport in One-Dimensional Objects

nanowires transport physics

2.3 Thermoelectric Phenomena in Zero-Dimensional Structures

What are the necessary changes that one has to take into account when dealing also with other extensive variable such as a temperature gradient

2.4 Density Functional Theory

What are the relevant parameters/quantities that we are interested to calculate ab initio Why we opted for a pseudopotential such as SIESTA rather than others (quantum espresso). Explain why calculating the isolated gas phase all-electron electronic structure is legit for a qualitative see how the electron density is distributed over the system.

Remember also to emphasise the difference from stricly and non-stricly localised functions used for computing the electronic properties: when the functions are stricly localised they are zero outside a given radius from the atom they are associated with and so the radius is a key quality determined parameter for the basis set

2.5 Non-Equilibrium Green Function Theory

This section is needed to explain formally the results from transiesta and TB-trans calculations

3

Instrumentations

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2.5	Non-Equilibrium Green Function Theory	22

3.1 Room Temperature DC Measurements

Describe here the probe station setup]

3.2 Dipstick DC Measurements

Describe here the dipstick used for both liquid N₂ and liquid He

3.3 Cryogenic DC Measurements

Describe here the delft box. Describe how the delft box modular structure enables to adapt two and multi-terminal DC measurements

3.4 Cryogenic AC Measurements

Describe here the delft box + lock-in setup and why it is so complicated to set it up

3.5 Thermoelectric Measurements

Describe here the setup used for the thermoelectric devices and why it needs a dedicated section Describe also here from a setup point of view which complications have raised during the setup (direct lines instead of dc lines, the need to use nano-D connectors and to design the a new pcb board).

4

Fabrication

Contents

3.1	Room Temperature DC Measurements	23
3.2	Dipstick DC Measurements	23
3.3	Cryogenic DC Measurements	23
3.4	Cryogenic AC Measurements	24
3.5	Thermoelectric Measurements	24

4.1 General Fabrication Procedure

The steps are well known: chip preparation, e-beam, development, metal deposition and lift-off, graphene transfer old process, graphenea graphene

4.1.1 Graphene Preparation and Wet Transferring

Initially the graphene was transferred on top of the substrate through the wet transfer method. The graphene was supplied by Graphene ain 5 x 5 cm2 rectangles, with a PMMA front-layer and a back-layer of partially removed graphene. Before transfer, the back-layer had to be fully removed via etching, and the remaining stack appropriately sized for the substrates for this work. The graphene arrived in a vacuum-sealed pack, which had to be transported to the cleanroom before

being opened and the following preparation steps initiated. Figure 6.2 depicts the main stages involved in preparing the graphene for transfe

4.2 Graphene Nanogaps

Here three methods have been exploited: standard feedback electroburning, gate-dependent electroburning, and lithographic patterning of graphene nanogap

4.3 Chip Design

Here the designs for the different chips that have been used. Give a proper credit to Dr. Alex Gee for the fabrication of thermoelectric and the common gate ones.

4.3.1 2T Design

How convenient and 'quick' is the usage of a backgate device. List, as usual, pro and cons

4.3.2 3T Design

Why the need to use 3T over 2T

Local Gates from Waterloo

Give proper credit

Interconnected Drain and Gate

The one I made and also made with the help of Alex

4.3.3 Thermoelectric Device

Common and different features of the thermoelectric

4.4 Deposition Process and Related Phenomna

Why it has been important to try different deposition techniques How this has been qualitatevely addressed.

5

Room Temperature Graphene Nanoribbon Single-Electron Transistor

Contents

4.1	General Fabrication Procedure	25
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4.3.2	3T Design	26
4.3.3	Thermoelectric Device	26
4.4	Deposition Process and Related Phenomna	26

5.1 Introduction

A brief intro why single-electron transistors with gnr are interesting A brief intro to what we are gonna do/measure in this chapter

5.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters)
Explained why the scarcity of characterisation checks is intrinsically scarce in molecular electronics.

5.3 Electrical Transport Measurements

What type of experiments have been performed and which tricks have been used in triton to do so. What are the evolution in temperature of the zero-bias, low-bias and high bias conductance. Look at the dependence in temperature and explain possible interpretations: why it is not a Luttinger Liquid (the gnr has a bandgap) and what possible phonon modes are responsible at some point for desctructing the

5.4 Summary

6

Chemical Doping of the Bandgap for Cove Graphene Nanoribbons

Contents

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6.1 Introduction

A brief intro on how why it is important to study this

6.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters)
Explained why the scarcity of characterisation checks i s intrinsically scarce in
molecular electronics.

6.3 *Ab-initio* Experiments

What DFT and NEGF have been used and for what.

6.4 Electrical Transport Measurements

Explain how the calculations have initially suggested the comparison between the base case and the methoxy groups, as they have the largest bandgap distance.

6.5 Summary

7

Thermocurrent Spectroscopy of covalently Functionalized Graphene Nanoribbons

Contents

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6.3	<i>Ab-initio</i> Experiments	29
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6.5	Summary	30

7.1 Introduction

A brief intro on how why it is important to study this what has been done.

7.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters)

Explained why the scarcity of characterisation checks is intrinsically scarce in molecular electronics.

7.3 Electrical Transport Measurements

Explain how the calculations have initially suggested the comparison between the base case and the methoxy groups, as they have the largest bandgap distance.

7.4 Summary

8

Transport Spectroscopy of Biaryl-linked Dy–Tb Porphyrin

Contents

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8.1 Introduction

A brief intro on how why it is important to study this

8.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters)

Explained why the scarcity of characterisation checks i s intrinsically scarce in
molecular electronics.

8.3 Electrical Transport Measurements

Explain how the calculations have initially suggested the comparison between the base case and the methoxy groups, as they have the largest bandgap distance.

8.4 Summary

9

Conclusions and Outlook

Contents

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8.4	Summary	34

9.1 Conclusions

Here the final conclusions

9.2 Outlook

Appendices

Cor animalium, fundamentum est vitæ, princeps omnium, Microcosmi Sol, a quo omnis vegetatio dependet, vigor omnis & robur emanat.

The heart of animals is the foundation of their life, the sovereign of everything within them, the sun of their microcosm, that upon which all growth depends, from which all power proceeds.

— William Harvey [66]



Review of Cardiac Physiology and Electrophysiology

Appendices are just like chapters. Their sections and subsections get numbered and included in the table of contents; figures and equations and tables added up, etc. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed et dui sem. Aliquam dictum et ante ut semper. Donec sollicitudin sed quam at aliquet. Sed maximus diam elementum justo auctor, eget volutpat elit eleifend. Curabitur hendrerit ligula in erat feugiat, at rutrum risus suscipit. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Integer risus nulla, facilisis eget lacinia a, pretium mattis metus. Vestibulum aliquam varius ligula nec consectetur. Maecenas ac ipsum odio. Cras ac elit consequat, eleifend ipsum sodales, euismod nunc. Nam vitae tempor enim, sit amet eleifend nisi. Etiam at erat vel neque consequat.

A.1 Anatomy

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A.2 Mechanical Cycle

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A.3 Electrical Cycle

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A.4 Cellular Electromechanical Coupling

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*The first kind of intellectual and artistic personality
belongs to the hedgehogs, the second to the foxes ...*

— Sir Isaiah Berlin [67]

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